Variation of Parameters

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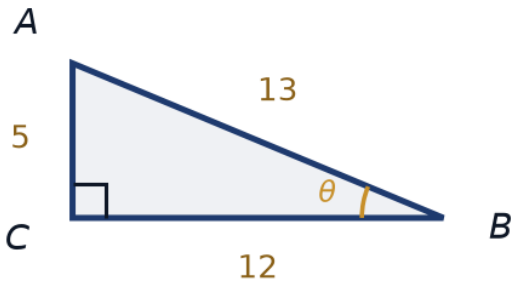
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: The Wronskian

For $y_1 = e^x$, $y_2 = e^{2x}$ (solutions of $y'' - 3y' + 2y = 0$), compute the Wronskian $W = y_1 y_2' - y_1' y_2$.

2 [Foundational] • Target: The Wronskian

For $y_1 = \cos x$, $y_2 = \sin x$, compute W.



3 [Foundational] • Target: The Wronskian

For $y_1 = e^{3x}$, $y_2 = xe^{3x}$ (solutions of a repeated-root equation), compute W.

4 [Foundational] • Target: The variation of parameters formula

State the variation of parameters formula for a particular solution of $y'' + p(x)y' + q(x)y = g(x)$, given independent solutions y_1, y_2 of the homogeneous equation with Wronskian W.

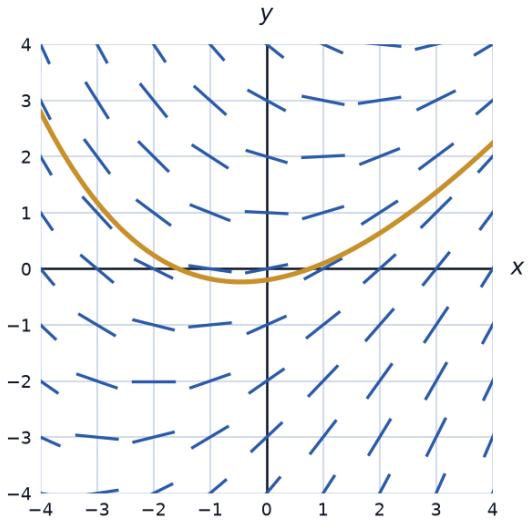
5 [Foundational] • Target: The variation of parameters formula

Explain why the ODE must first be written in standard form $y'' + p(x)y' + q(x)y = g(x)$ (leading coefficient 1) before applying the variation of parameters formula.

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6 [Foundational] • Target: A forcing term with no undetermined-coefficients trial form

Solve $y'' + y = \tan x$ for $-\frac{\pi}{2} < x < \frac{\pi}{2}$ using variation of parameters.



7 [Foundational] • Target: A forcing term with no undetermined-coefficients trial form

Solve $y'' + y = \sec x$ for $-\frac{\pi}{2} < x < \frac{\pi}{2}$ using variation of parameters.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: A repeated root, with forcing dividing by x

Solve $y'' - 2y' + y = \frac{e^x}{x}$ $(x > 0)$ using variation of parameters.

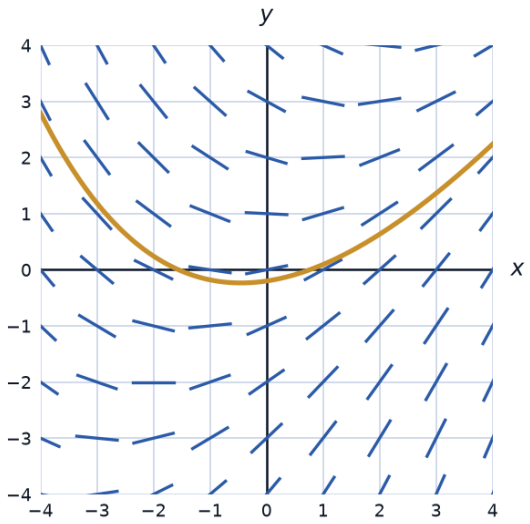
9 [Intermediate] • Target: A repeated root, with forcing dividing by x

Solve $y'' - 4y' + 4y = \frac{e^{2x}}{x^2}$ $(x > 0)$ using variation of parameters.

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10 [Intermediate] • Target: Comparing with undetermined coefficients

Solve $y'' - y = e^{2x}$ by variation of parameters, then check the particular solution using undetermined coefficients.



11 [Intermediate] • Target: Comparing with undetermined coefficients

Solve $y'' - y = 4$ (a constant forcing) by variation of parameters, and check against the obvious undetermined-coefficients guess $y_p = A$.

12 [Intermediate] • Target: When to prefer each method

For $y'' + 4y = x^3 - 2x$, which method is more efficient: undetermined coefficients or variation of parameters? Explain.

13 [Intermediate] • Target: When to prefer each method

For $y'' + y = \csc x$ (for $0 < x < \pi$), which method must be used, and why?

14 [Intermediate] • Target: Solving $y''+y=\csc x$ in full

Complete the solution of $y'' + y = \csc x$ ($0 < x < \pi$) started above, using $y_1 = \cos x$, $y_2 = \sin x$, $W = 1$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

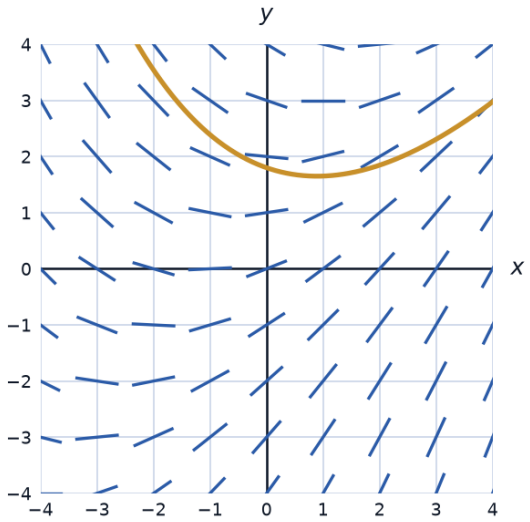
15 [Challenging] • Target: Solving $y''+y=\csc x$ in full • Cross-Lesson: 3.01 Second-Order Linear Homogeneous Equations

Verify (by checking the coefficient of the leading term only) that $y_p = -x\cos x$ contributes a term of the correct order when substituted into $y'' + y$.

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16 [Challenging] • Target: Variation of parameters with a shifted independent variable • Cross-Lesson: 4.01 Nonhomogeneous Equations & Undetermined Coefficients

For the Cauchy–Euler-type substitution used in some ODE courses, $y_1 = x$, $y_2 = x^{-1}$ solve $x^2y'' + xy' - y = 0$ for $x > 0$. Rewrite this equation in the standard form $y'' + p(x)y' + q(x)y = g(x)$ needed for variation of parameters when the right side is $x^2 \cdot 3x = 3x^3$ (i.e., solve $x^2y'' + xy' - y = 3x^3$).



17 [Challenging] • Target: Variation of parameters with a shifted independent variable • Cross-Lesson: 1.01 Separable Equations

Continue the problem above: with $y_1 = x$, $y_2 = x^{-1}$, compute $W = y_1y_2' - y_1'y_2$.

18 [Challenging] • Target: Variation of parameters with a shifted independent variable • Cross-Lesson: 2.01 Linear First-Order Equations & the Integrating Factor

Finish solving $x^2y'' + xy' - y = 3x^3$ $(x > 0)$ using $y_1 = x$, $y_2 = x^{-1}$, $W = -2x^{-1}$, and $g(x) = 3x$ (from the two questions above).

19 [Challenging] • Target: Variation of parameters with a shifted independent variable • Cross-Lesson: 3.01 Second-Order Linear Homogeneous Equations

Check the particular solution $y_p = \frac{3}{8}x^3$ found above by substituting directly into $x^2y'' + xy' - y = 3x^3$.

20 [Challenging] • Target: A summary comparison • Cross-Lesson: 4.01 Nonhomogeneous Equations & Undetermined Coefficients

List two advantages of variation of parameters over undetermined coefficients.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: The Wronskian Determinant

What is the definition of the Wronskian $W(y_1, y_2)$ of two differentiable functions $y_1(x)$ and $y_2(x)$?

a. $W(y_1, y_2) = y_1y_2' - y_1'y_2 = \det[[y_1, y_2], [y_1', y_2']]$

b. $W(y_1, y_2) = y_1y_2 + y_1'y_2'$

c. $W(y_1, y_2) = y_1'y_2 - y_1y_2'$

d. $W(y_1, y_2) = \frac{y_1}{y_2}$

22 [Foundational] • Target: The Wronskian Determinant

Compute the Wronskian of $y_1 = \cos x$ and $y_2 = \sin x$.

a. 1

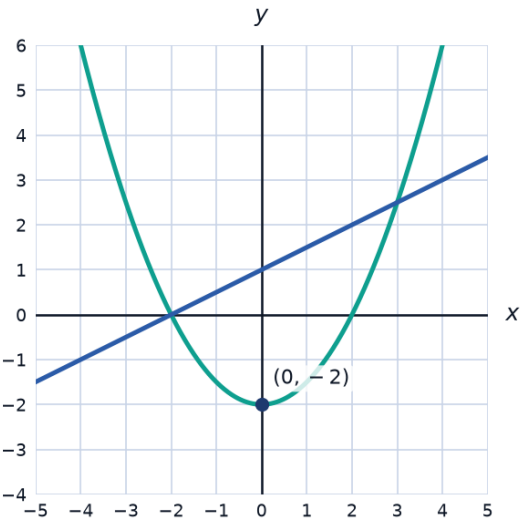
b. 0

c. -1

d. $\cos(2x)$

23 [Foundational] • Target: The Wronskian Determinant

What does $W(y_1, y_2) \neq 0$ guarantee about solutions y_1 and y_2 to a homogeneous linear ODE?



a. They are linearly independent and form a fundamental set of solutions

b. They are orthogonal

c. They are periodic

d. One must be zero

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24 [Foundational] • Target: The Variation of Parameters Formula

For the second-order equation $y'' + P(x)y' + Q(x)y = g(x)$, what is the variation of parameters formula for y_p?

a. $y_p = -y_1 \int \frac{y_2 g}{W} dx + y_2 \int \frac{y_1 g}{W} dx$

b. $y_p = y_1 \int \frac{y_2 g}{W} dx - y_2 \int \frac{y_1 g}{W} dx$

c. $y_p = \int \frac{g(x)}{W} dx$

d. $y_p = -y_2 \int \frac{y_1}{W} dx$

25 [Foundational] • Target: The Variation of Parameters Formula

What must the coefficient of y'' be before applying the variation of parameters formula?

a. Exactly 1 (must be in standard monic form)

b. Any non-zero constant

c. Zero

d. Positive

26 [Foundational] • Target: The Variation of Parameters Formula

For $y'' + y = \sec x$, where $y_1 = \cos x$, $y_2 = \sin x$, what is $u_1' = -\frac{y_2 g}{W}$?

a. $-\tan x$

b. $-\sin x$

c. -1

d. $-\cos x$

27 [Foundational] • Target: Solving Advanced Nonhomogeneous Equations

Find $u_1 = \int (-\tan x) dx$ for the problem $y'' + y = \sec x$.

a. $\ln|\cos x|$

b. $-\ln|\cos x|$

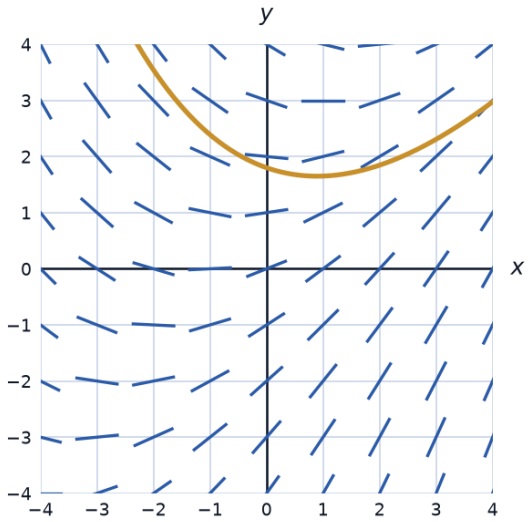
c. $\sec^2 x$

d. $-\cos x$

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28 [Foundational] • Target: Solving Advanced Nonhomogeneous Equations

Find $u_2' = \frac{y_1 g}{W}$ for $y'' + y = \sec x$ and evaluate u_2.



a. $u_2 = x$

b. $u_2 = \sin x$

c. $u_2 = \cos x$

d. $u_2 = \tan x$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Solving Advanced Nonhomogeneous Equations

Assemble the particular solution $y_p = u_1 y_1 + u_2 y_2$ for $y'' + y = \sec x$.

a. $y_p = \cos x \ln|\cos x| + x \sin x$

b. $y_p = \cos x + x \sin x$

c. $y_p = \ln|\cos x| + x$

d. $y_p = \sin x \ln|\cos x| + x \cos x$

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30 **[Intermediate]** • **Target: Comparison of Nonhomogeneous Methods**

What is the major advantage of Variation of Parameters over Undetermined Coefficients?

a. It works for any continuous forcing function $g(x)$, not just polynomials, exponentials, or sines/cosines

b. It never requires any integration

c. It works without knowing the homogeneous solution y_c

d. It always produces algebraic rather than transcendental answers

31 **[Intermediate]** • **Target: Comparison of Nonhomogeneous Methods**

What is the main drawback of Variation of Parameters compared to Undetermined Coefficients?

a. The integrals $\int \frac{y_i g}{W} dx$ may be difficult or impossible to evaluate in closed elementary form

b. It only works on first-order equations

c. It gives only approximate numerical solutions

d. It cannot handle initial value problems

32 **[Intermediate]** • **Target: Comparison of Nonhomogeneous Methods**

What is Abel's formula for the Wronskian of solutions to $y'' + P(x)y' + Q(x)y = 0$?

a. $W(x) = Ce^{-\int P(x)dx}$

b. $W(x) = Ce^{\int Q(x)dx}$

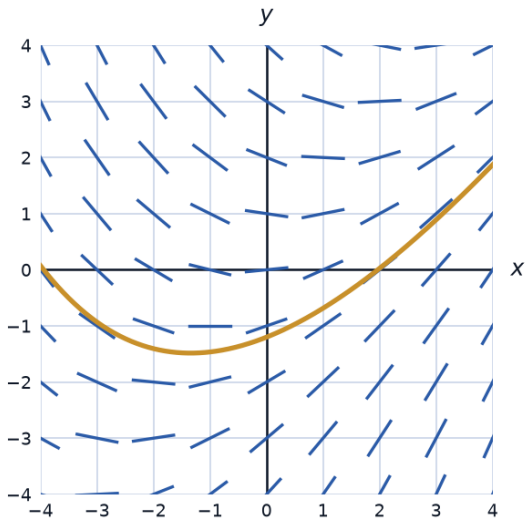
c. $W(x) = C(P(x) + Q(x))$

d. $W(x) = Cx$

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33 [Intermediate] • Target: Comparison of Nonhomogeneous Methods

Consider the second-order linear nonhomogeneous differential equation $y'' - y = \sin x$. If $y_p(x) = A\cos x + B\sin x$ is a particular solution, find the values of the constants A and B.



a. $A = 0, B = \frac{1}{2}$

b. $A = -\frac{1}{2}, B = 0$

c. $A = 0, B = -\frac{1}{2}$

d. $A = \frac{1}{2}, B = 0$

34 [Intermediate] • Target: The Wronskian

Solve $y'' + 3y' + 2y = -6e^{-4x} - 2$, $y(0) = -1$, $y'(0) = 1$. Find $y(1)$.

a. $-1 - e^{-1} - e^{-4} + \frac{2}{e^2}$

b. $-\frac{2}{e^2} + e^{-4} + e^{-1} + 1$

c. $-2 - \frac{2}{e} - \frac{2}{e^4} + \frac{4}{e^2}$

d. $-\frac{1}{2} - \frac{1}{2e} - \frac{1}{2e^4} + e^{-2}$

35 [Intermediate] • Target: The variation of parameters formula

Solve $y'' - 2y' - 3y = -21e^{-4x} - 3$, $y(0) = -4$, $y'(0) = 0$. Find $y(1)$.

a. $-1 + e^{-4} + \frac{2}{e} + 2e^3$

b. $-4e^3 - \frac{4}{e} - \frac{2}{e^4} + 2$

c. $-e^3 - e^{-1} - \frac{1}{2e^4} + \frac{1}{2}$

d. $-2e^3 - \frac{2}{e} - e^{-4} + 1$

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36 [Intermediate] • Target: A forcing term with no undetermined-coefficients trial form

Solve $y'' + 4y' + 4y = e^{-2x}$, $y(0) = 0$, $y'(0) = 0$. Find $y(2)$.

a. $2e^4$

b. e^{-4}

c. $4e^{-4}$

d. $2e^{-4}$

37 [Intermediate] • Target: A repeated root, with forcing dividing by x

Solve $y'' - y' - 6y = -6e^{-3x} - 6$, $y(0) = -4$, $y'(0) = -4$. Find $y(1)$.

a. $-1 + e^{-3} + e^{-2} + 3e^3$

b. $-3e^3 - e^{-2} - e^{-3} + 1$

c. $-6e^3 - \frac{2}{e^2} - \frac{2}{e^3} + 2$

d. $-\frac{3e^3}{2} - \frac{1}{2e^2} - \frac{1}{2e^3} + \frac{1}{2}$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Comparing with undetermined coefficients • Cross-Lesson: 2.01 Linear First-Order Equations & the Integrating Factor

Solve $y'' - y' - 2y = -10e^{4x} - 4x$, $y(0) = -7$, $y'(0) = -6$. Find $y(1)$.

a. $-e^4 - 3e^2 - \frac{2}{e} + 1$

b. $-1 + \frac{2}{e} + 3e^2 + e^4$

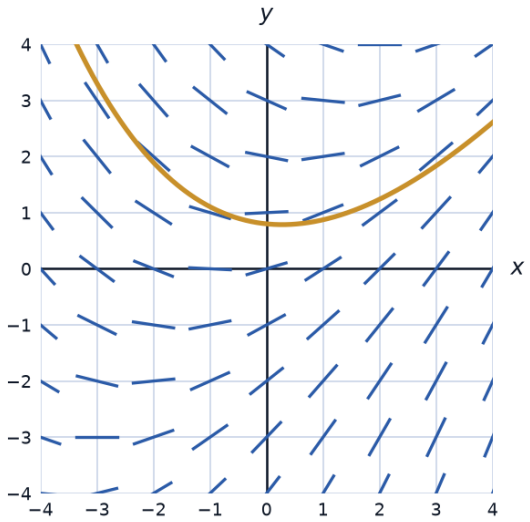
c. $-2e^4 - 6e^2 - \frac{4}{e} + 2$

d. $-\frac{e^4}{2} - \frac{3e^2}{2} - e^{-1} + \frac{1}{2}$

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39 [Challenging] • Target: When to prefer each method • Cross-Lesson: 3.01 Second-Order Linear Homogeneous Equations

Solve $y'' + 2y' - 3y = -21e^{4x} - 9x$, $y(0) = 2$, $y'(0) = 8$. Find $y(1)$.



a. $-3e - 5 + \frac{2}{e^3} + e^4$

b. $-2e^4 - \frac{4}{e^3} + 10 + 6e$

c. $-\frac{e^4}{2} - e^{-3} + \frac{5}{2} + \frac{3e}{2}$

d. $-e^4 - \frac{2}{e^3} + 5 + 3e$

40 [Challenging] • Target: Solving $y''+y=\csc x$ in full • Cross-Lesson: 4.01 Nonhomogeneous Equations & Undetermined Coefficients

Solve $y'' - y' - 6y = 6e^{-3x} + 6$, $y(0) = 2$, $y'(0) = -12$. Find $y(1)$.

a. $-e^3 - 1 + e^{-3} + \frac{3}{e^2}$

b. $-\frac{3}{e^2} - e^{-3} + 1 + e^3$

c. $-2e^3 - 2 + \frac{2}{e^3} + \frac{6}{e^2}$

d. $-\frac{e^3}{2} - \frac{1}{2} + \frac{1}{2e^3} + \frac{3}{2e^2}$

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41 **[Challenging]** • **Target: Variation of parameters with a shifted independent variable** • **Cross-Lesson: 1.01 Separable Equations**

Solve $y'' - y' - 2y = -10e^{-3x} - 2$, $y(0) = 2$, $y'(0) = 4$. Find $y(1)$.

a. $-e^2 - 1 - e^{-1} + e^{-3}$

b. $-\frac{1}{e^3} + e^{-1} + 1 + e^2$

c. $-\frac{2}{e^3} + \frac{2}{e} + 2 + 2e^2$

d. $-\frac{1}{2e^3} + \frac{1}{2e} + \frac{1}{2} + \frac{e^2}{2}$

42 **[Challenging]** • **Target: A summary comparison** • **Cross-Lesson: 2.01 Linear First-Order Equations & the Integrating Factor**

Solve $y'' - y = -15e^{-4x} - 1x$, $y(0) = -2$, $y'(0) = 0$. Find $y(1)$.

a. $-3e - e^{-4} + \frac{2}{e} + 1$

b. $-1 - \frac{2}{e} + e^{-4} + 3e$

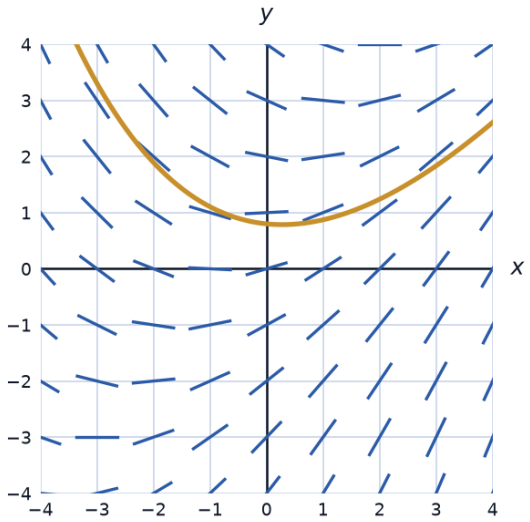
c. $-6e - \frac{2}{e^4} + \frac{4}{e} + 2$

d. $-\frac{3e}{2} - \frac{1}{2e^4} + e^{-1} + \frac{1}{2}$

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43 [Challenging] • Target: The Wronskian Determinant • Cross-Lesson: 3.01 Second-Order Linear Homogeneous Equations

Solve $y'' - y = -15e^{-4x} - 1$, $y(0) = 0$, $y'(0) = 2$. Find $y(1)$.



a. $-1 - e^{-1} + e^{-4} + e$

b. $-e - e^{-4} + e^{-1} + 1$

c. $-2e - \frac{2}{e^4} + \frac{2}{e} + 2$

d. $-\frac{e}{2} - \frac{1}{2e^4} + \frac{1}{2e} + \frac{1}{2}$

44 [Challenging] • Target: The Variation of Parameters Formula • Cross-Lesson: 4.01 Nonhomogeneous Equations & Undetermined Coefficients

Solve $y'' + 10y' + 25y = 3e^{-5x}$, $y(0) = 3$, $y'(0) = -13$. Find $y(1)$.

a. $\frac{13}{2e^5}$

b. $-\frac{13}{2e^5}$

c. $\frac{13}{e^5}$

d. $\frac{13}{4e^5}$

45 [Challenging] • Target: Solving Advanced Nonhomogeneous Equations • Cross-Lesson: 1.01 Separable Equations

Solve $y'' + 8y' + 16y = 3e^{-4x}$, $y(0) = -2$, $y'(0) = 11$. Find $y(2)$.

a. $\frac{10}{e^8}$

b. $-\frac{10}{e^8}$

c. $\frac{20}{e^8}$

d. $\frac{5}{e^8}$

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